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Editor-in-Chief
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Dear Editor,

I'm writing to submit my paper, "Institutional Observability Under Scaled AI Governance: Deployment-Scale Capacity Degradation in Human Oversight Pipelines," for consideration in Technology in Society. I'm a student at SM Shetty International School in Mumbai, and this is my first time submitting to an academic journal. I hope that's okay.

The paper is about something I started noticing while reading about AI governance last year. Everyone talks about bias and fairness and transparency—and those are important, obviously—but I kept thinking: even if you solve all of that, what happens when there are just too many decisions for humans to actually review? Like, if a company deploys an AI system that flags 10,000 cases a day, and they have 5 human reviewers, it doesn't matter how good their compliance dashboard looks. The math doesn't work.

I ended up building a model for this. I called it oversight saturation. The basic idea is that human review capacity plateaus (it's bounded by how fast people can read and think) while AI decision volume scales linearly with deployment. Above a certain threshold—I found $\rho_c = 0.60$ based on the human factors literature—review quality doesn't just degrade gradually. It collapses. And the compliance dashboards that institutions use to prove they're doing oversight? They can't detect this collapse because they count artifacts, not actual review depth.

The thing I'm most proud of is SEDI—the State-Estimation Degradation Index. It's a number between 0 and 1 that tells you how much you can actually trust an institution's oversight based on public records alone. You don't need access to their internal systems. You just need three things: how fast they're generating compliance artifacts, how long cases take to resolve, and what their appeal rate looks like. I validated it with Monte Carlo simulations (120 runs, nothing fancy, just numpy on my laptop) and calibrated it against real cases from Vanderbilt's academic integrity pipeline and the Peterson

Health Technology Institute's work on prior authorization.

I know this paper is a bit unusual. It's got queueing theory and state-space models in it, which you don't normally see from a high school student. But I think the core argument is actually pretty simple: if you scale AI decisions faster than you scale the humans who review them, oversight breaks. Not because anyone is negligent. Because the architecture makes it inevitable. And right now, our governance frameworks don't have a way to measure when this is happening.

I think Technology in Society is the right place for this because the paper sits between a few different fields. It's not a pure CS paper about queueing algorithms. It's not a pure policy paper either. It's about what happens to governance institutions when technology scales past their capacity to understand it. That feels like exactly the kind of conversation this journal has been having, especially with the recent work on AI ethics and algorithmic accountability.

A few practical things:

The manuscript is about 6,500 words with 5 figures and 1 table. The simulation code is all up on GitHub (https://github.com/SilverCoin256/oversight-saturation-sedi) with a README that explains how to run everything. I've included all the declarations the journal requires—AI disclosure, ethics statement, data availability, CRediT, conflict of interest. There's no funding to declare since I did this on my own.

I've suggested five reviewers below who I think would understand what I'm trying to do. They're all people whose work I cited and genuinely admire. I'd be nervous about them reading my paper, honestly, but I think they'd give fair feedback.

If there's anything wrong with the formatting or if I missed something in the submission guidelines, please let me know. I read the guide for authors carefully but this is my first time and I might have messed something up.

Thank you for taking the time to read this. I know getting a submission from a high schooler is probably not what you expected today.

Respectfully,

Shaurya Gupta

Suggested Reviewers

1. **Dr. Inioluwa Deborah Raji** — University of California, Berkeley
She basically defined what internal AI auditing looks like in practice. Her FAccT paper on closing the AI accountability gap is one of the main reasons I got interested in this topic. She'd know immediately if my arguments hold up.

2. **Dr. Sarah Lebovitz** — University of Michigan
Her work on how practitioners shift from real verification to post-hoc rationalization under load is exactly the behavioral pattern my model predicts. I'd really value her read on whether the depth degradation function captures what she's observed empirically.

3. **Dr. Meg Young** — Cornell Tech / Data & Society
She works on algorithmic accountability in the public sector and has written about audit infrastructure design. The SEDI index is meant to be something public-sector auditors could actually use, so her perspective would be incredibly helpful.

4. **Dr. Michael Veale** — University College London
His expertise in GDPR, AI regulation, and governance-by-design is directly relevant to the governance principles I derive in the last section. He'd catch it if I'm making claims about regulatory frameworks that don't hold up.

5. **Dr. Frank Pasquale** — Cornell University
His work on algorithmic accountability and the black box society is foundational. The accountability routing architecture I map out in Figure 3 is essentially a formalization of concerns he's been raising for years about who actually holds power in automated systems.